MILITARY SPECIFICATION

CARTRIDGE, 105MM, TP-T, M393A1 CONVERSION

This specification is approved for use by the U.S. Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

This specification covers the conversion of the High Explosive M393A2 105MM HEP-T Cartridge to a Target Practice Cartridge M393A1 105MM TP-T. It applies only to disassembled cartridge components previously rejected during renovation of the M393A2.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-S-13812	-	Steel Plate, Wrought Homogeneous, for Ammunition Testing
MIL-A-48078	-	Ammunition, Standard Quality Assurance Provisions, General Specifications for

FSC 1315

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-P-50980 - Charge, Propelling, for Use with 105MM Cartridges, M393A2 HEP-T; M393A1 TP-T; M416 WP-T; M467 TP-T, Loading, Assembling and Packing

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-129 - Marking for Shipment and Storage

MIL-STD-414 - Sampling Procedures and Tables for Inspection by Variables

MIL-STD-1167 - Ammunition Data Card

MIL-STD-1168 - Lot Numbering of Ammunition

MIL-STD-1169 - Packaging, Packing and Marking of Inert Ammunition Components

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND (ARRADCOM)

PRODUCT AND PACKAGING DRAWINGS

9335943 - Cartridge, 105MM, TP-T, M393A1 (Conversion) Assembly and Marking Diagram

8836004 - Box, Packing, Ammunition for Cartridge 105MM, HEP-T, TP-T, WP-T, and Dummy for Gun M68 in Fiber Container

8836005 - Container, Ammo, Fiber M451 for Cartridge 105MM, HEP-T, TP-T, WP-T, and Dummy for Gun M68

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INSPECTION EQUIPMENT DRAWINGS

8828098 - Gage, Profile and Alignment

8828099 - Gage, Alignment Check

8828100 - Gage, Wear Limit Check

TECHNICAL MANUALS

TM-9-1300-250 Ammunition Maintainance

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Material. Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Complete round. The Cartridge shall comply with all requirements specified in Drawing 9335943 and with all requirements specified in applicable specifications.

3.3 Components. Projectile metal parts shall be free from cracks, separations, burrs or other surface defects.

3.4 Cartridge case crimp (bullet pull). The crimped cartridge case shall provide a bullet pull of 4,500 pounds minimum, to 9,000 pounds maximum, when tested as specified in 4.6.2.

3.5 Proving ground. The cartridge shall comply with the following requirements when tested as specified in 4.6.3.

3.5.1 Muzzle velocity. The muzzle velocity of the cartridge when fired at 70 degrees fahrenheit (21.1° centigrade) shall not be more than 2430 feet per second (fps) (740.7 meters per second (mps)) nor less than 2370 fps (722.4 mps) and the lot standard deviation shall not be greater than 10.5 fps (4.0 mps).

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3.5.2 Tracer functioning. The tracer shall ignite and there shall be visible evidence of burning to the 2000 meter target.

3.5.3 Target impact dispersion. The probable error shall not be greater than 0.20 mils horizontal and 0.21 mils vertical (see 4.5.4.5.3) when tested in accordance with 4.6.3.

3.5.4 Safety. There shall be no hangfires or indications of metal parts failure in the gun bore or in flight, prior to impact, during any of the proving ground tests conducted.

3.6 Workmanship.

3.6.1 General. The cartridge and all components shall be free of cracks, burrs, chips, dirt, grease, (including thread preservative), rust, or other foreign material. The cleaning method shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents. The cartridge shall not be damaged in any manner during the loading operation.

3.6.2 Painting and marking. The painting and marking shall be restored by the assembling contractor on any components furnished to him painted and marked, when such paintings and markings have beem marred or obliterated. (Bare spots in exterior painting, other than marking, are allowable provided that they total not more than 1/4 inch and that the maximum dimension of any individual bare spot does not exceed 3/16 inch).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)

b. Quality conformance inspection (see 4.5)

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4.3 Test methods and procedures. Unless otherwise specified, all inspections shall be performed in accordance with the test methods and procedures specified in 4.6.

4.4 First article inspection

4.4.1 Submission. The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with provisions of 4.4.2. The first article sample for the loading facility shall consist of 90 cartridges for ballistic testing as listed in Table I. The sample shall be manufactured from the same materials, equipment, processes and procedures as will be used in the regular conversion of cartridges. All parts and materials, including packaging and packing shall be the same as used for regular production or conversion and shall be obtained from the same source of supply. Submission of the manufacturing process and related process controls to be utilized by the manufacturer to assure a quality product shall be made to the contracting officer. Detailed operation sheets for LAP prepared in accordance with the general format of Figure 4-5 TM9-1300-250 shall also be submitted to the contracting officer. Upon approval of the first article sample, the process controls submitted at the time of the sample shall constitute the minimal controls required to assure a quality product.

4.4.2 Inspections to be performed. The first article cartridges submitted in accordance with 4.4.1 will be subjected by the Government to any or all requirements of the applicable drawings and to the inspections of this specification. All inspections are to be performed 100 percent unless otherwise specified. The ballistic tests will be conducted in accordance with the procedures of 4.6.3 and with the sample sizes and phases as listed in Table I.

4.4.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon failure of any assembly, component or test specimen in the sample to comply with any of the stated requirements.

4.4.4 First article inspection.

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63527 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.4	Cartridge, 105MM, TP-T, M393A1	1 OF 1	9335943
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	$70^{\circ}F \pm 2.5^{\circ}F$	30			
	Velocity-Accuracy				
	Muzzle Velocity				4.6.3.1
	Target Impact Dispersion				4.6.3.1
	Tracer Functioning				4.6.3.3
	Flight				4.6.3.4
	Safety				4.6.3.11
	$145^{\circ}F - 5^{\circ}F$	30			
	Safety				4.6.3.11
	Tracer Functioning (Info Only)				4.6.3.3
	Muzzle Velocity (Info Only)				4.6.3.1
	Target Impact Dispersion (Info Only)				4.6.3.1
	$-50^{\circ}F + 5^{\circ}F$	30			
	Safety				4.6.3.11
	Tracer Functioning (Info Only)				4.6.3.3
	Muzzle Velocity (Info Only)				4.6.3.1
	Target Impact Dispersion (Info Only)				4.6.3.1

NOTES:

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4.5 Quality conformance inspection.

4.5.1 Inspection lot formation. Inspection lots shall comply with the provisions of MIL-A-48078. The integrity of each lot prior to and following conversion shall be maintained. Replacement components, assembled by the loading plant into a converted lot, shall be from not more than one lot interfix. Moreover, a lot of each component (i.e. base plug, packing seal, felt, paper disc, etc.) shall be used in consecutive order of lot acceptance, until it is exhausted before another lot is introduced into production.

4.5.2 Examination. One-hundred percent examination shall be performed for all critical, major and minor defects when so specified. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein using the sampling plan of MIL-STD-105, Level II. All non-conforming material shall be rejected. See MIL-A-48078. In the event of a dispute, four power (4x) magnifiction shall be used in performing visual inspections for such defects as burrs and foreign matter. The roughness comparison specimens prescribed by ASA-B46.1 shall be used as a basis for surface roughness determination.

4.5.3 Table II classfications of defects and tests.

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TABLE II
QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63527 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.5.3.1	Projectile, Prior to Loading Inert Filler				9335945
					NEXT HIGHER ASSEMBLY
					9335944
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
201	Projectile body color incorrect		100%	3.2	Visual
202	Foreign matter in charge cavity		0.40%	3.2	Visual
Minor					
301	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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TABLE II
QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63527 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
3.5.3.2	Projectile Prior to Assembly of Base Plug	1 OF 1	9335944
			NEXT HIGHER ASSEMBLY
			9335943

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major					
101	Packing Seal Missing		0.40%	3.2	Visual
102	Depth to filler incorrect		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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TABLE II
QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63527 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.5.3.3	Plug, Base				9323957
					NEXT HIGHER ASSEMBLY
					9335944
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Presence of cracks		0.40%	3.1	Visual
102	Damage to thread		0.40%	3.1	Visual
103	Damage to wrench holes		0.40%	3.1	Visual
Minor					
201	Protective coating inadequate or damaged		0.65%	3.1	Visual
202	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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TABLE II
QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63527 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.5.3.4	Packing Seal		9323812
			NEXT HIGHER ASSEMBLY
			9335944

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Thickness		0.65%	3.1	Gage
202	Outside diameter		0.65%	3.1	Gage
203	Inside diameter		0.65%	3.1	Gage
204	Concentricity of inside diameter with outside diameter		0.65%	3.1	Gage
205	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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TABLE II
QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63527 (AR)

PARAGRAPH	TITLE			SHEET 1 OF 1	DRAWING NUMBER
4.5.3.5	Ring				8860609
					NEXT HIGHER ASSEMBLY
					9335944
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Thickness		0.65%	3.1	Gage
202	Material torn, damaged or cracked		0.65%	3.1	Visual
203	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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TABLE II
QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63527 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.5.3.6	Primer, Prior to Assembly to Cartridge Case				8847424
					NEXT HIGHER ASSEMBLY
					10521648
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
101	One or more flash holes missing		100%	3.1	Visual
Major					
101	Check test for deterioration			3.1	4.5.4.1/Test
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES: Required only if replacing primers in cartridge case

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TABLE II
QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63527 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.5.3.7	Cartridge Case Prior to Insertion of Projectile				10521648
					NEXT HIGHER ASSEMBLY
					9335943
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Electrical resistance		100%	3.2	4.6.1
102	Propelling charge assembled with foil at wrong end		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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TABLE II
QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63527 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.5.3.8	Projectile Loading Assembly				9335944
					NEXT HIGHER ASSEMBLY 9335943
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Weight		100%	3.2	Gage (SMTE)
102	Diameter after retouching		0.40%	3.2	Gage
103	Base plug fails to withstand reapplication of minimum assembly torque		0.40%	3.2	Gage
104	Tracer fails to withstand reapplication of minimum assembly torque		0.40%	3.2	Gage
105	Ring missing or damaged		0.40%	3.2	Visual
106	Tracer plug and disc assembly damaged		0.40%	3.2	Visual
Minor					
201	Length, max		0.65%	3.2	Gage
202	Protective coating with bare spots		0.65%	3.2	Visual
203	Marking incorrect misleading, or unidentifiable including incorrect color code		0.65%	3.2	Visual
204	Evidence of poor workmanship		0.65%	3.6	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63527 (AR)

PARAGRAPH	TITLE		SHEET	DRAWING NUMBER
4.5.3.9	Cartridge Assembly		1 OF 1	9335943
				NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Profile and alignment		100%	3.2	4.6.4/Gage
102	Gap between cartridge case and rotating band, max.		0.40%	3.2	Gage
103	Cartridge case crimp			3.2	4.6.2/Test
104	Distance from rear face of cartridge case to top of electrode, max.		0.40%	3.2	Gage
105	Primer stake missing		0.40%	3.2	Visual
Minor					
201	Marking misleading or unidentifiable including incorrect color codes		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.6	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63527 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.5.3.10	Disc		9335917
			NEXT HIGHER ASSEMBLY
			9335944

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter		0.40%	3.1	Gage
102	Thickness		0.40%	3.1	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63527 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.5.3.11	Cushion				9335918
					NEXT HIGHER ASSEMBLY
					9335944
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter		0.40%	3.1	Gage
102	Thickness		0.40%	3.1	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63527 (AR)

PARAGRAPH	TITLE			SHEET 1 OF 1	DRAWING NUMBER
4.5.3.12	Unsealed Fiber Container				8836005
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Any packing component missing or improperly assembled		0.40%	3.2	Visual
Minor					
201	Cartridge cannot be removed by hand		0.65%	3.2	Manual
202	Container cap cannot be removed by hand		0.65%	3.2	Manual
203	Glue or asphalt on cartridge		0.65%	3.2	Visual
204	Evidence of poor workmanship		0.65%	3.6	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63527 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.5.3.13	Sealed Fiber Container				8836005
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Sealing tape incomplete		0.40%	3.2	Visual
102	Container cut or damaged through all asphalt impregnated layers		0.40%	3.2	Visual/Manual
103	Metal end loose or distorted		0.40%	3.2	Visual/Manual
Minor					
201	Sealing tape tab length, min.		0.65%	3.2	Gage
202	Contents loose		0.65%	3.2	Manual
203	Gap between cover and body of container exceeding 1/8 inch		0.65%	3.2	Visual
204	Cuts, scuffs or gouges in outer layers		0.65%	3.2	Visual
205	Marking misleading or unidentifiable		0.65%	3.2	Visual
206	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63527 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.5.3.14	Sealed Wood Packing Box				8836004
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Box damaged exposing contents		0.40%	3.2	Visual
102	Hardware or strapping missing, broken or loose		0.40%	3.2	Visual/Manual
103	DOD symbol missing or incorrect		0.40%	3.2	Visual
Minor					
201	Strapping or hardware improperly assembled		0.65%	3.2	Visual/Manual
202	Handle missing or insecure		0.65%	3.2	Visual/Manual
203	Car seal missing, unsealed or improperly positioned		0.65%	3.2	Visual
204	Marking incorrect, misleading or unidentifiable		0.65%	3.2	Visual
205	Evidence of poor workmanship		0.65%	3.6	Visual

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4.5.4 Testing.

4.5.4.1 Check test for deterioration of primers. (See applicable primer specification) Major defect. If the total elapsed time between original acceptance of any primer lot and the assembly of that lot into complete round cartridges exceeds two years, or if the primers have been subjected to adverse conditions, however brief, at any time since previous tests, the applicable lot shall be subjected to and must satisfactorily pass the check test specified in the applicable primer specification immediately before the primer lot is assembled into the cartridges. The test shall be performed by the contractor on assemblies selected by the Government inspector at the facility assembling the complete round cartridges. (See 6.4)

4.5.4.2 Cartridge case crimp (bullet pull). (See 3.4 and dwg. 10521648) Major defect. The sampling plan for this test shall be in accordance with MIL-STD-414, Section B, Part II, Table B3, Sample Size Code Letter I and an AQL of 1.5%. The method used shall be in accordance with 4.6.2 using equipment in accordance with 4.5.5. The samples chosen for this test shall have successfully passed the profile and alignment test of 4.6.4. Prior to the start of each shift of production, an additional sample of 3 cartridges from each crimping machine used shall be tested for bullet pull. If any of these samples provides a bullet pull not in accordance with the requirements of 3.4, the crimping machine represented by the sample shall not be utilized until additional samples of 3 cartridges, as required, can meet the applicable requirement following adjustments or repairs to the affected crimping machine, as applicable. All samples shall be gaged for profile and alignment prior to testing.

4.5.4.3 Projectile weight. (see 3.1 and dwg. 9335944) Major defect. In order to determine whether the variation in projectile weight falls within acceptable limits, a random sample of 25 projectiles of the first hour's production of each cartridge lot shall be weighed and recorded. The accuracy of the scale used shall be to the nearest 0.01 $\pm$ 0.005 pound. A mean and standard deviation shall be calculated for this sample. If the standard deviation exceeds 0.075 lbs., the contractor shall cease propellant charge loading of the cartridge lot, determine the cause of the large variation in weight and submit his findings to the contracting officer for evaluation and lot disposition. In addition, the projectiles selected for sampling for minor defects in 4.5.3.8 shall also be weighed and the actual weights recorded.

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4.5.4.4 Propellant weight. If the projectile mean weight is within 25.00 ± 0.10 pounds, then the cartridge lot shall be assembled using the authorized charge weight, as determined by a Government proving ground, for a 25.00 pound projectile. If the samples mean weight falls within 24.80 + 0.09 or 25.20 - 0.09 pounds, then the authorized propellant charge weight, as determined by a Government proving ground, for a 24.80 or 25.20 pound projectile, respectively, shall be used.

4.5.4.5 Proving ground tests. The following tests shall be performed as specified in 4.6.3.

4.5.4.5.1 Muzzle velocity. (See 3.5.1) - Major Defect.

4.5.4.5.1.1 First article test. The sampling plan for this test shall be in accordance with MIL-STD-414, Section B, Part II, Table B3, Code letter J. An AQL = 1.0% shall be used. In addition, if the velocity standard deviation of the sample cartridges multiplied by the factor 0.77 exceeds the lot requirements, the lot shall be rejected.

4.5.4.5.1.2 Lot acceptance. The sampling plan shall be in accordance with MIL-STD-414, Section B, Part II, Table B3, Code letter G. An AQL of 1.0 percent shall be used. In addition, if the velocity standard deviation of the sample cartridges multiplied by the factor 0.65 exceeds the applicable lot requirements, the lot shall be rejected.

4.5.4.5.2 Tracer functioning (see 3.5.2) Major Defect.

4.5.4.5.2.1 First article sample. If, during the velocity-accuracy phase of 4.6.3.1, three or more tracers fail to comply with the applicable requirement, the first article sample shall be rejected.

4.5.4.5.2.2 Lot acceptance. All cartridges fired in the ballistic tests prescribed by the specification shall be observed for tracer functioning. If, during the velocity-accuracy phase of 4.6.3.1, two or more tracers fail to function to the target, the lot represented by the ballistic sample shall be rejected.

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4.5.4.5.3 Target impact dispersion. (see 3.5.3) Major Defect. The sample cartridges used in the test for muzzle velocity shall be observed as specified in 4.6.3.1 for Target Impact Dispersion. If the pooled Probable Error sample calculated in accordance with 6.7 exceeds PE_2 the lot shall be rejected.

TABLE III

PROBABLE ERRORS

Sample Size	PE_1	PE_2	PE_3
15 Horizontal	0.20	0.30	0.41
15 Vertical	0.21	0.31	0.43
30 Horizontal	0.20	0.27	0.33
30 Vertical	0.21	0.28	0.35

NOTE: The sampling plan, shown in Table III above, will provide an A (alpha) risk of 0.01 and a B (beta) risk of 0.10 to discriminate between PE_1 (the value associated with the 1.0 percent producer's risk) and PE_3 (the value associated with the 10.0 percent consumer's risk).

4.5.4.5.4 Extreme temperature firing.

4.5.4.5.4.1 First article test. Thirty cartridges shall be tested at $-50^{o}F + 5^{o}F$ and $145^{o}F - 5^{o}F$ for muzzle velocity, target impact dispersion and tracer functioning as specified in 4.6.3.2.

4.5.4.5.4.2 Lot acceptance. Ten cartridges from each lot shall be fired for safety at $+145^{o}F - 5^{o}F$. Tracer functioning to impact shall be recorded for information. Testing shall be conducted in accordance with 4.6.3.2.

4.5.4.5.5 Safety. (see 3.5.4) Critical Defect. All cartridges fired in the proving ground tests shall be observed for hangfires or break-up of metal parts. If any cartridge fails to comply with the applicable requirements the lot shall be rejected.

4.5.5 Inspection equipment. The inspection equipment required by the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.5.3.1 . The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See section 6 of MIL-A-48078.

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4.6 Test methods and procedures.

4.6.1 Resistance. Resistance shall be measured between the primer electrode and the cartridge case using an electrical circuit which limits the current through the primer to 20 milliamperes max., with the primer electrode as one contact point and the cartridge case as the other.

4.6.2 Cartridge case crimp (bullet pull). The cartridge shall be tested in an appropriate tensile testing machine to determine the security of the crimp of the cartridge case to the projectile. The load shall be applied in an axial direction with the rate of travel of the testing machine's head from 0.100 to 0.125 inch per minute.

4.6.3 Proving ground test procedures. Samples to be ballistically tested as prescribed herein shall be fired in an M68 gun. The ammunition shall be conditioned to the appropriate temperature until all propellant is stabilized at a uniform temperature prior to firing, which normally requires a period of 24 hours, minimum. Muzzle velocity shall be recorded for all rounds, including calibration ammunition and spotters. In those phases in which muzzle velocity will be taken for information only, firing cadence of a group will not be disrupted because of velocity equipment failure. All samples shall be observed for safety. Sufficient warmers shall be fired by the proving ground to assure temperature stabilization of the weapon prior to the starting of any phase of this ballistic test. Maximum chamber pressure shall be recorded for all calibration rounds.

4.6.3.1 Velocity-accuracy phase.

4.6.3.1.1 The data generated in the velocity-accuracy phase will primarily be used for determination of acceptance from a target impact disperion/muzzle velocity standpoint, and will require the firing of calibration rounds to establish a velocity correction. The firing sequence will be five calibration rounds, fifteen test rounds, and five calibration rounds. This phase will be fired using an M60 or M60A1 tank with a new to mid-life gun tube (see 6.6). A fixed mount may be used providing that the procedure of 4.6.3.1.4, or its equivalent, is able to remove the backlash from the fire control system.

4.6.3.1.2 Target distance is 2000 meters and target size is 6 meter by 6 meter minimum .

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4.6.3.1.3 The ammunition fired in this phase shall be preconditioned to the temperature of 70°F ± 2.5°F (21.1°C ± 1.4°C).

4.6.3.1.4 After loading and locking a round in the chamber, the gun shall be traversed off the target center, dropped down below the target center, traversed to the opposite side of the target center, elevated to the target center, and traversed onto the target center. The distance traversed and dropped should be sufficient to remove the backlash from the system. If the sight is above or beyond the target center, the sequence will be repeated without reversal of gun movement direction.

4.6.3.1.5 Spotter rounds will be used to assure the weapon is zeroed to assure engagement of the target center. A minimum of three consecutive target hits within a circle of a one mil radius from the target center by spotters fired at the same elevation and deflection are required. An aiming point or calibration rounds may be used for this purpose. Once the gun has been spotted on the target for a shot group, the elevation and deflection of the weapon shall remain the same for that shot group.

4.6.3.1.6 The test samples shall be fired in five round groups (or occasions) with a cadence of no less than three minutes between rounds and not more than five minutes between rounds. If more than a thirty minute lapse occurs between group firings or if there is an indication of a trend in the previous group's target impacts to move away from the center of the target, the weapon shall be re-zeroed on target center. A cadence of less than three minutes will be allowed, providing it can be maintained until all test and calibration rounds are fired.

4.6.3.1.7 Target impacts (including spotters) shall be measured to the nearest one-half inch (one centimeter) in both the horizontal and vertical distance from target center.

4.6.3.1.8 Vertical and horizontal hit pattern results shall be expressed as probable errors in mils and shall be recorded on the firing record. No more than one lot acceptance test shall be performed on each target. This test phase shall be conducted using a new to mid-life gun tube (see 6.6 for tube wear definitions).

4.6.3.2 Extreme temperature firing. The cartridges selected for temperature conditioning shall be fired in a mid-life tube (see 6.6).

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4.6.3.3 Tracer functioning. Cartridges fired during all tests in which target impact dispersion data is taken shall be observed for tracer functioning to the target. The distance over which the tracer burns shall be recorded in increments of 500 meters (e.g. failure, 500 meters, 1000 meters).

4.6.3.4 Flight. An observer will be located at an elevated point behind the firing point with an unobstructed view of the line of flight to the target and shall document projectile flight. Abnormalities of flight will be described (corkscrewing, yaw, etc.) as well as target impact information (hit target, short by 100 meters, left target, etc.). If a round misses the target with no abnormality of flight, a statement to that effect will also be made.

4.6.3.5 Wind velocity. Wind velocity and gusts shall be measured at the firing point and at the target prior to and after the velocity/target dispersion test group, and at the firing point prior to firing each round of the group. The velocity/dispersion portion of the ballistic test shall not be fired if the nominal surface wind velocity and surface wind gusts exceed 10 knots (18.5) km/hr and 5 knots (9.25 km/hr) respectively at either location. Only that component of the surface wind normal to the line of fire will be considered. In addition, peak surface wind velocity (nominal plus gusts) normal to the line of fire is not to exceed 10 knots (18.5 km/hr) during the firing of any shot group during the velocity/dispersion phase.

4.6.3.6 Target misses. Target miss coordinates will be estimated and recorded for all test phases along with pertinent flight observations.

4.6.3.7 Retest. If for any reason the proving ground considers that test conditions have detrimentally affected the test results, 15 additional cartridges, as required, shall be tested. A lot shall be retested whenever the proving ground deems it necessary, or whenever any of the following conditions exist:

a. The muzzle velocity correction factor computed from the average muzzle velocity of the calibration rounds exceeds 30 fps.

b. The standard deviation of the muzzle velocity of the calibration rounds, multiplied by the factor 0.65 exceeds 10.5 fps.

c. Two or more velocities are lost when firing the test of 4.6.3.1.

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d. The surface wind velocity normal to the gun target line exceeds 10 knots during the firing of any test group.

e. The cadence between any two successive rounds in a test group exceeds 5 minutes.

f. A change was made in the azimuth or elevation of the weapon within a test group.

A retest for probable error (PE) is not necessary when target impact dispersion test results are acceptable.

4.6.3.8 Target miss. Cartridge lots which have one target miss during the target impact disperion phase of the proving ground tests described herein shall be retested if both of the following conditions are met:

a. One or more of the retest criteria (para 4.6.3.7 above) are met.

b. The pooled PE of the remaining rounds meet the requirements of para. 4.5.4.4.3.

If either condition is not met, the lot shall be rejected. Two or more target misses by the test samples shall automatically reject the lot. Results of retests for target impact dispersion shall be referred to the contracting officer for disposition.

4.6.3.9 Calibration cartridge loading. The calibration projectiles shall be inert loaded and assembled at the proving ground so as to assure an assured weight of 24.80 ± .01 pounds. The projectiles and components shall be assembled into cartridges in accordance with drawing 9335943.

4.6.3.10 Camera coverage. Adequate photographic equipment shall be set up to assure satisfactory coverage of each test projectile's integrity and flight from the gun muzzle to the first one hundred feet of flight, minimum, for each phase of this ballistic acceptance test. A smear camera at 100 ft or less from the muzzle is acceptable coverage.

4.6.3.11 Safety. All cartridges fired in proving ground tests shall be observed for hang-fires, or break-up of projectile metal parts.

4.6.4 Profile and alignment check. Each crimped cartridge's profile and alignment shall be verified by a gage corresponding to

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dwg. 8828098, which is to be horizontal ($0^{o} \pm 2.5^{o}$). The cartridge shall be inserted in accordance with the requirements of dwg. 9335943. The profile and alignment gage shall not be utilized as an alignment tool.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. The cartridge shall be packaged in accordance with dwg. 8836005 and level A of MIL-STD-1169.

5.2 Packing. Cartridges in containers shall be packed in accordance with 8836004 and level A of MIL-STD-1169.

5.3 Marking. Wooden boxes and fiber containers shall be marked and sealed in accordance with dwgs. 8836004 and 8836005 and level A of MIL-STD-1169.

6. NOTES.

6.1 Ordering Data. See MIL-A-48078 (PA)

In addition to the requirements of MIL-A-48078, procurement documents shall specify the following as a minimum:

a. The title, number and date of this specification

b. That data cards be prepared for each lot in accordance with MIL-STD-1167.

6.2 Submission of inspection equipment designs for approval. See 6.2.3 of MIL-A-48078. Submit equipment designs as required to: Commander, ARRADCOM, ATTN: DRDAR-QAT-I, Dover, NJ 07801.

6.3 Submission of test data and ammunition data cards. In addition to the normal distribution of records, one copy of all ballistic test data and ammunition data cards shall be forwarded to each of the following: Commander, ARRADCOM ATTN: DRDAR-LCU-CV and DRDAR-QAT-B, Dover, NJ 07801; and Commander, ARRCOM, ATTN: DRSAR-QAD, Rock Island, IL 61299.

6.4 Cost of check test. The contracting officer shall arrange for the contractor to be reimbursed for the expense incurred in the performance of the check tests. The tests shall be conducted at Government expense without cost to the contractor who loaded the assemblies or to the contractor assembling the

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cartridge, and shall not constitute a basis for rejection against either contractor, except where deterioration has occurred as direct result of carelessness in handling, storage, etc., permitted while the primer lot was under the jurisdiction of either contractor.

6.5 Firing record information.

6.5.1 Target impact plots, meteorological data, star gage and pull over gage measurements shall also be included in all firing records.

6.5.2 Firing records shall include the serial numbers of all weapons and fire control components, wind direction and velocity and weapon to target azimuth. Stock component lots of warmers and spotters shall also be included.

6.5.3 Each round fired during the test shall be identified in the sequence of firing. For each round fired, the record shall contain deflection, elevation, time of firing, muzzle velocity, tracer functioning horizontal and vertical coordinates of impact, and flight characteristics.

NOTE: Chamber pressure shall be recorded for the calibration rounds.

6.6 Tube wear definitions.

6.6.1 Primary wear. Wear measured in a vertical plane 25.25 inches from the rear face of the tube.

6.6.2 New tube. A tube with no greater than 0.015 inch primary land wear.

6.6.3 Mid-life tube. A tube in which the primary land wear is between 0.016 inch and 0.029 inch inclusively.

6.6.4 Worn tube. a tube in which:

(1) The primary land wear is over 0.030 inches.

(2) The secondary land wear is between 70 percent and 140 percent of the primary land wear.

(3) The secondary groove wear is between 80 percent and 190 percent of the primary land wear.

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6.7 Method for calculating pooled probable error (PE).

6.7.1 First article test. The individual PE results of the six groups shall be pooled according to the following method:

$$\text{Pooled PE} = \left[\sum_{i=1}^{6} (N_i - 1)(PE_i)^2 \Big/ \sum_{i=1}^{6} (N_i - 1) \right]^{1/2}$$

Where PE_i and N_i are probable error value and sample size, respectively, for each of the six groups of five samples fired for target impact dispersion in the first article ballistic test.

6.7.2 Lot acceptance. The individual PE results of the three groups shall be pooled according to the following method:

$$\text{Pooled PE} = \left[\frac{(N_A-1)\ (PE_A)^2 + (N_B-1)\ (PE_B)^2 + (N_C-1)\ (PE_C)^2}{(N_A-1) + (N_B-1) + (N_C-1)} \right]^{1/2}$$

Where PE_A and N_A, PE_B and N_B, PE_C and N_C are the Probable Error and Sample Size of each of the groups, respectively.

6.8 Calibration inspection lots. Calibration inspection lots are defined as lots of projectiles or sub-assemblies which have been inspected under conditions of continuous production by a process requiring periodic calibration. The number of items inspected between calibrations shall constitute each calibration inspection lot.

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6.9 Proving Ground Test Summary.

Test	Sample Size First Article	Sample Size Lot Acceptance	Requirements	Sampling Test Paragraph	Procedures Paragraph
Velocity-Accuracy	30	15	Muzzle Velocity	4.5.4.5.1	4.6.3.1
			Target Impact Dispersion	4.5.4.5.3	4.6.3.1
70°F ± 2.5°F			Tracer Functioning	4.5.4.5.2	4.6.3.3
			Flight		4.6.3.4
			Safety	4.5.4.5.5	4.6.3.11
Safety 145°F -5°F	30	10	Safety	4.5.4.5.5	4.6.3.11
			Muzzle Velocity	4.5.4.5.1	4.6.3.1
			Target Impact Dispersion	4.5.4.5.3	4.6.3.1
			Tracer Functioning	4.5.4.5.2	4.6.3.3
Safety -50°F+5°F	30	0*	Safety	4.5.4.5.5	4.6.3.11
			Muzzle Velocity	4.5.4.5.1	4.6.3.1
			Target Impact Dispersion	4.5.4.5.3	4.6.3.1
			Tracer Functioning	4.5.4.5.2	4.6.3.3

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